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# Customer Credit Data Analysis using Python

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Course : Bachelor of technology
Internship Domain : Data Analytics
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Introduction

This project analyzes customer credit data using Python. The dataset contains
information about customers such as age, credit limit, and other attributes.
The aim of this project is to explore the dataset and understand patterns
using data analysis techniques.

Python libraries such as Pandas, Matplotlib, and Seaborn are used to perform
data analysis and visualization. Various charts and statistical techniques
are used to understand the data.
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Tools and Technologies Used

• Python
• Google Colab
• Pandas
• Matplotlib
• Seaborn
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("credit_card.csv")
df.head()
```

	Sno	Customer	Age	City	Credit_Type	Limit	Company	Segment
0	1	A1	1	BANGALORE	Gold	INR 500,000.00	C1	Self Employed
1	2	A2	36	CALCUTTA	Silver	INR 100,000.00	C2	Salaried_MNC
2	3	A3	12	COCHIN	Platimum	INR 10,000.00	C3	Salaried_Pvt
3	4	A4	46	BOMBAY	Platimum	INR 10,001.00	C4	Govt
4	5	A5	70	BANGALORE	Platimum	INR 10,002.00	C5	Normal Salary

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(100, 8)
```

```
df.columns
```

```
Index(['Sno', 'Customer', 'Age', 'City', 'Credit_Type', 'Limit', 'Company',
       'Segment'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   Sno          100 non-null    int64 
 1   Customer     100 non-null    object
 2   Age          100 non-null    int64 
 3   City         100 non-null    object
 4   Credit_Type  100 non-null    object
 5   Limit        100 non-null    object
 6   Company      100 non-null    object
 7   Segment      100 non-null    object
dtypes: int64(2), object(6)
memory usage: 6.4+ KB
```

```
df.describe()
```

	Sno	Age
count	100.000000	100.000000
mean	50.500000	40.170000
std	29.011492	23.584387
min	1.000000	1.000000
25%	25.750000	20.500000
50%	50.500000	41.000000
75%	75.250000	61.250000
max	100.000000	78.000000

```
df.isnull().sum()
```

	0
Sno	0
Customer	0
Age	0
City	0
Credit_Type	0
Limit	0
Company	0
Segment	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['Limit'] = df['Limit'].str.replace("INR ", "", regex=False)
df['Limit'] = df['Limit'].str.replace(",", "", regex=False)
df['Limit'] = df['Limit'].astype(float)
```

```
df['Limit'].dtype
```

```
dtype('float64')
```

```
df.shape
```

```
(100, 8)
```

```
round(df['Age'].mean(), 2)
```

```
np.float64(40.17)
```

```
print("Youngest:", df['Age'].min())
print("Oldest:", df['Age'].max())
```

```
Youngest: 1
Oldest: 78
```

```
round(df['Limit'].mean(), 2)
```

```
np.float64(284700.61)
```

```
df['Credit_Type'].value_counts()
```

Credit_Type	count
Gold	38
Platimum	32
Silver	30

dtype: int64

```
df['Segment'].value_counts()
```

Segment	count
Govt	29
Self Employed	23
Normal Salary	22
Salaried_Pvt	13
Salaried_MNC	13

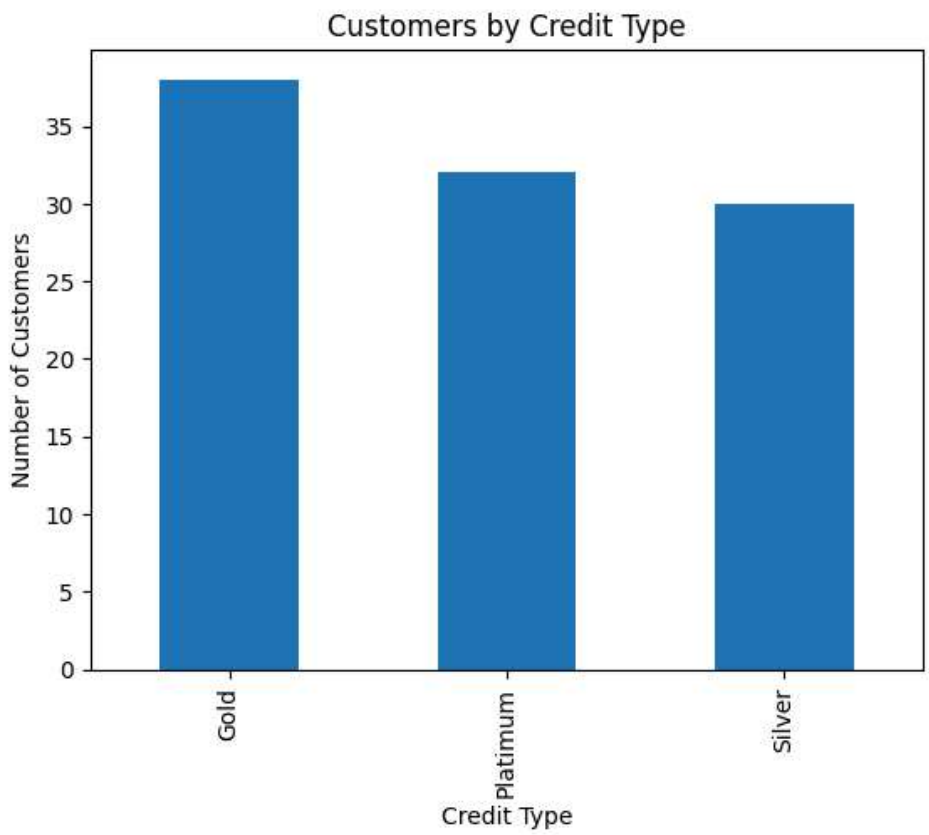
dtype: int64

```
df.groupby('Segment')['Limit'].mean()
```

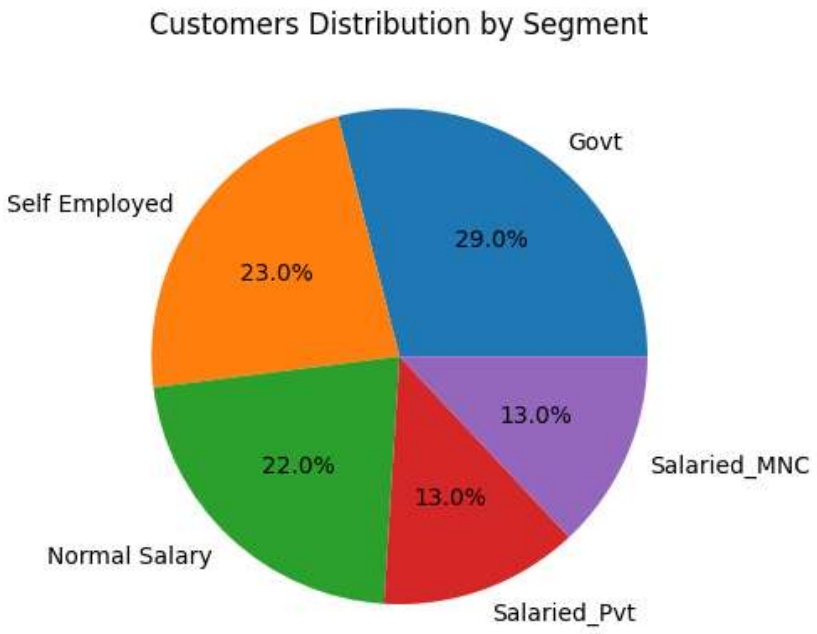
Segment	Limit
Govt	308276.482759
Normal Salary	283637.318182
Salaried_MNC	270769.692308
Salaried_Pvt	270769.538462
Self Employed	271739.652174

dtype: float64

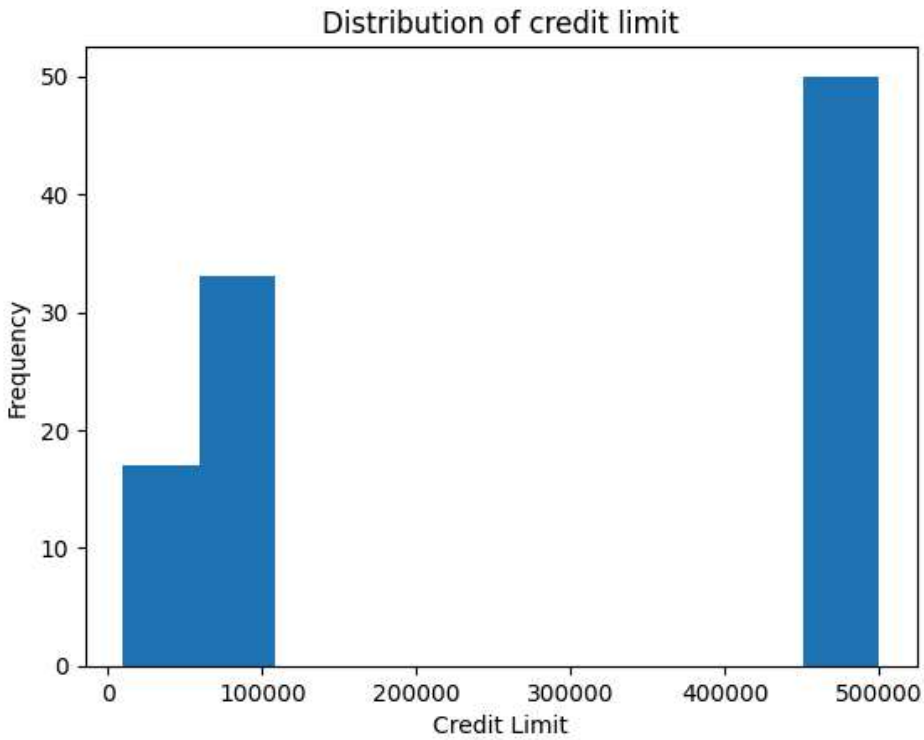
```
df['Credit_Type'].value_counts().plot(kind='bar')
plt.title("Customers by Credit Type")
plt.xlabel("Credit Type")
plt.ylabel("Number of Customers")
plt.show()
```



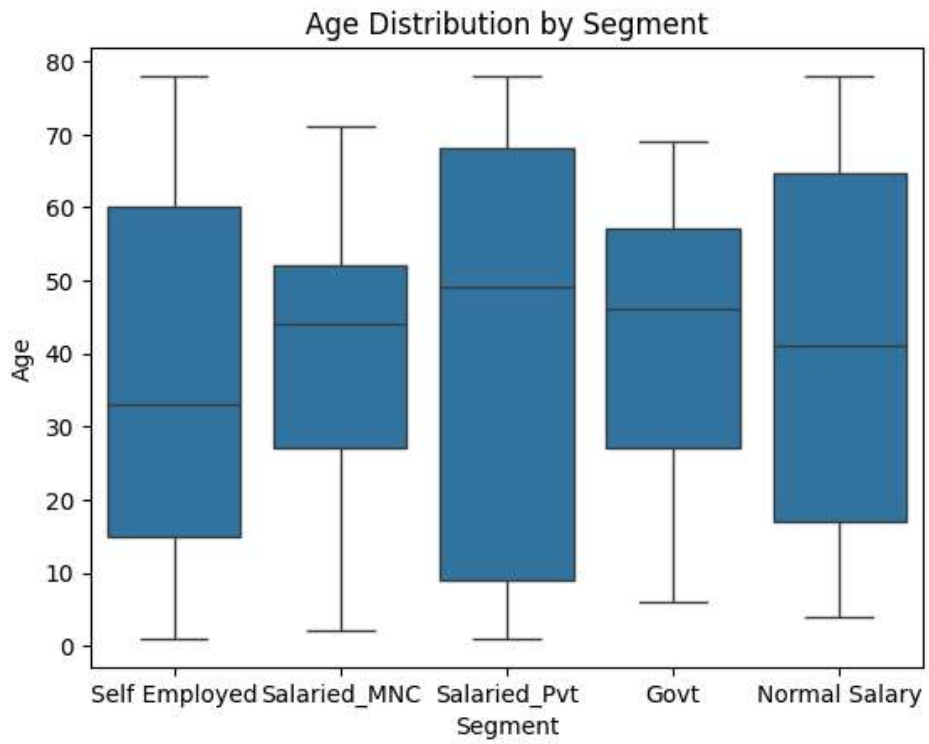

```
df['Segment'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Customers Distribution by Segment")
plt.ylabel("")
plt.show()
```



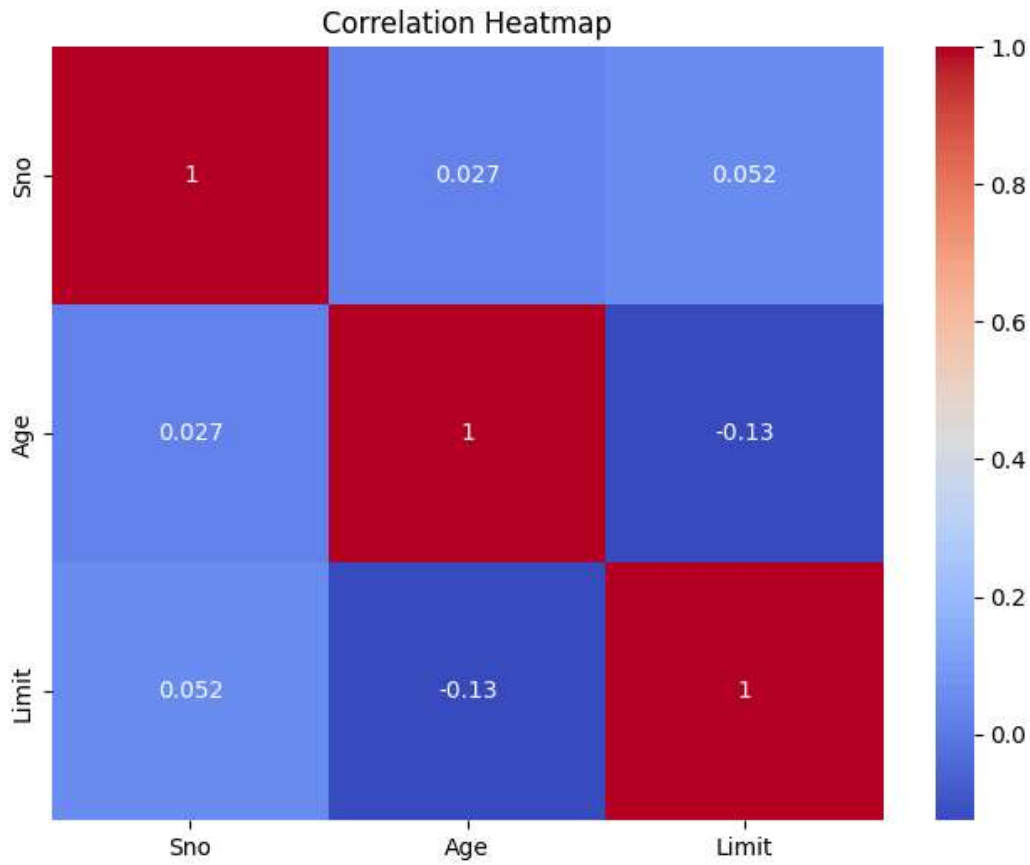

```
plt.hist(df['Limit'], bins=10)
plt.title("Distribution of credit limit")
plt.xlabel("Credit Limit")
plt.ylabel("Frequency")
plt.show()
```




```
sns.boxplot(x='Segment', y='Age', data=df)
plt.title("Age Distribution by Segment")
plt.show()
```




```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



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## Key Insights

- The dataset contains 100 customer records with 8 attributes.

- Data visualization techniques such as bar charts, pie charts,
  histograms, and boxplots were used to understand the data distribution.

- Credit limit values vary significantly among customers.

- The correlation heatmap shows that most variables have weak
  relationships with each other.

- Visualization helps identify patterns and trends in the dataset.
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## Conclusion

This project analyzed customer credit data using Python and
data analysis libraries such as Pandas, Matplotlib, and Seaborn.
Different visualization techniques were used to understand the
distribution and patterns in the dataset.

Correlation analysis was also performed to examine relationships
between numeric variables. The project demonstrates how data
analytics techniques can be applied to extract useful insights
from raw data.
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